

Integrated Experiments – Problem Statement

Experiment 1: Decision Tree Classification

Objective: Implement and evaluate a Decision Tree classifier on a given dataset using Python / any ML library.

(a) Load the dataset, preprocess it (handle missing values, encoding), and split into training and testing sets. Display the first 5 rows and data types.

(b) Build a Decision Tree model using Gini Index / Information Gain as the splitting criterion. Print the tree structure and identify the root node with justification.

(c) Evaluate the model using accuracy score, confusion matrix, and classification report. Comment on performance and list at least two misclassified instances.

Experiment 2: Reinforcement Learning – Q-Learning

Objective: Implement Q-Learning for a simple 4×4 Grid World environment and observe the agent's learned policy.

(a) Define states and actions (Up/Down/Left/Right), with rewards +10 Goal, -1 each step, -5 obstacle. Initialize the Q-table with zeros and state α , γ , ϵ .

(b) Run Q-Learning for at least 100 episodes. Record Q-values at Episodes 1, 50, and 100 for two representative states. Show how the Q-table evolves.

(c) Extract and display the final learned policy. Plot cumulative reward per episode and justify convergence behaviour.